

Industrial Internship Report

Project Title: Banking Information System

Prepared by: Nemmani Bhargav Harsha

Internship: upskill Campus / UCT

1. Table of Contents

1. Preface
2. Introduction
3. About UCT
4. About upskill Campus
5. Objectives
6. Problem Statement
7. Existing & Proposed Solution
8. System Design
9. High Level Design
10. Low Level Design
11. Interfaces
12. Performance Testing
13. Test Cases
14. Learnings
15. Future Scope

1. Preface
2. Introduction
3. About UCT
4. About upskill Campus
5. Objectives
6. Problem Statement
7. Existing & Proposed Solution
8. System Design
9. High Level Design
10. Low Level Design
11. Interfaces
12. Performance Testing
13. Test Cases
14. Learnings
15. Future Scope

2. Preface

This internship provided practical exposure to software development using Core Java. I designed and implemented a Banking Information System using Object-Oriented Programming concepts. The project strengthened my problem-solving, programming, testing, and documentation skills.

This internship provided practical exposure to software development using Core Java. I designed and implemented a Banking Information System using Object-Oriented Programming concepts. The project strengthened my problem-solving, programming, testing, and documentation skills.

3. Introduction

The Banking Information System is a console-based Java application that automates common banking operations. The application uses classes, objects, encapsulation, ArrayList, and exception handling to manage customer accounts and transactions.

The Banking Information System is a console-based Java application that automates common banking operations. The application uses classes, objects, encapsulation, ArrayList, and exception handling to manage customer accounts and transactions.

4. About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt. Ltd. (UCT) works in digital transformation, IoT, cloud computing, machine learning, industrial automation, and software development. The internship introduced students to industry-oriented software development.

UniConverge Technologies Pvt. Ltd. (UCT) works in digital transformation, IoT, cloud computing, machine learning, industrial automation, and software development. The internship introduced students to industry-oriented software development.

5. About upskill Campus

upskill Campus provides career-focused internships, industrial projects, and skill development programs. It bridges academic learning with industry requirements.

upskill Campus provides career-focused internships, industrial projects, and skill development programs. It bridges academic learning with industry requirements.

6. Objectives

- Learn Core Java
 - Apply OOP concepts
 - Develop a complete project
 - Practice GitHub version control
 - Improve software documentation skills
-
- Learn Core Java
 - Apply OOP concepts
 - Develop a complete project
 - Practice GitHub version control
 - Improve software documentation skills

7. Problem Statement

Manual banking operations are time-consuming and susceptible to errors. The objective is to develop a Java application that automates account creation, deposits, withdrawals, balance enquiry, and transaction history while providing a simple user interface.

Manual banking operations are time-consuming and susceptible to errors. The objective is to develop a Java application that automates account creation, deposits, withdrawals, balance enquiry, and transaction history while providing a simple user interface.

8. Existing and Proposed Solution

Existing systems often involve manual processing or fragmented software. The proposed Java application centralizes customer account management with automated calculations and transaction recording.

<https://github.com/BhargavHarshaN/upskillcampus/blob/main/BankingInformationSystem.java>

Report GitHub Link: <PASTE PDF LINK>

Existing systems often involve manual processing or fragmented software. The proposed Java application centralizes customer account management with automated calculations and transaction recording.

Code GitHub Link: <PASTE JAVA FILE LINK>

Report GitHub Link: <PASTE PDF LINK>

9. System Design

Modules:

- Account Creation
- Deposit
- Withdraw
- Balance Enquiry
- Account Details
- Transaction History

Data Flow:

User → Menu → BankingInformationSystem → BankAccount → Output

Modules:

- Account Creation
- Deposit
- Withdraw
- Balance Enquiry
- Account Details
- Transaction History

Data Flow:

User → Menu → BankingInformationSystem → BankAccount → Output

10. High Level Diagram

[Insert High-Level Diagram Here]

User → Menu → BankingInformationSystem → BankAccount → Transaction
Processing → Output

[Insert High-Level Diagram Here]

User → Menu → BankingInformationSystem → BankAccount → Transaction
Processing → Output

11. Low Level Diagram

[Insert Low-Level Diagram Here]

Main() → Menu → Switch Case → Search Account → Perform Operation → Display Result

[Insert Low-Level Diagram Here]

Main() → Menu → Switch Case → Search Account → Perform Operation → Display Result

12. Interfaces

Console Interface:

1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. Account Details
6. Transaction History
7. Exit

Console Interface:

1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. Account Details
6. Transaction History
7. Exit

13. Performance Test

The application was tested with multiple accounts and transactions. The system produced correct balances and transaction histories while maintaining fast execution and low memory usage.

The application was tested with multiple accounts and transactions. The system produced correct balances and transaction histories while maintaining fast execution and low memory usage.

14. Test Cases

1. Create Account – Pass
2. Deposit – Pass
3. Withdraw – Pass
4. Invalid Account – Pass
5. Balance Enquiry – Pass
6. Transaction History – Pass

1. Create Account – Pass
2. Deposit – Pass
3. Withdraw – Pass
4. Invalid Account – Pass
5. Balance Enquiry – Pass
6. Transaction History – Pass

15. My Learnings

This project enhanced my understanding of Java programming, OOP, collections, exception handling, software testing, debugging, GitHub, and technical documentation.

This project enhanced my understanding of Java programming, OOP, collections, exception handling, software testing, debugging, GitHub, and technical documentation.

16. Future Scope

Future enhancements include MySQL integration, Swing/JavaFX GUI, login authentication, password encryption, fund transfer, account deletion, and online banking services.

Future enhancements include MySQL integration, Swing/JavaFX GUI, login authentication, password encryption, fund transfer, account deletion, and online banking services.